Harish G

ML / Data Science Engineer | Forecasting in Production, Embeddings & Retrieval

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Profile

I report what a model actually does, including when that is disappointing. The forecasting system I own ships with a certification: median error of 16.6%, and only 18.7% of series inside the client's requested 10% band. I could have narrowed the intervals until that number looked right. I did not. Everything I ship is scored out of sample against a seasonal-naive baseline that is allowed to win.

Core Skills

Languages	Python · SQL · JavaScript · Java
Modelling	LightGBM · Prophet · scikit-learn · PyCaret · Quantile Regression · Conformal Prediction · Feature Engineering · QLoRA Fine-Tuning · Keras/LSTM
Evaluation	WAPE · MASE · WMAPE · MAE / RMSE / R^2 · Out-of-Sample Splits · Seasonal-Naive Baselines
Serving	FastAPI · joblib · Model Hot-Reload · Retrain Pipelines · Docker & Compose · Streamlit · GitLab CI
AI / LLM	RAG · MiniLM & text-embedding-004 · pgvector · Neo4j Graph Enrichment · Whisper · Gemini · Ollama
Data	PostgreSQL · Supabase/pgvector · MongoDB · Neo4j · Redis

Experience

Software Developer, CortexCraft.ai Jan 2026 – Present

- Built and maintain the voice-agent platform (589 of 626 commits on the SaaS platform, 67 of 75 on the core engine), including the per-agent retrieval that grounds each agent in its own documents and the post-call extraction that turns transcripts into structured fields.
- Design the output guardrails and tool-calling contracts that constrain model output: extracted data is coerced to its declared type before anything downstream trusts it.
- Lead author (172 of 192 commits) on the multilingual survey platform: Google speech-to-text and text-to-speech behind a config-driven registry covering 24 locales, with fuzzy and semantic fallback matching so imperfect speech still resolves to an option.

Freelance Web & App Developer (Self-Employed) 2023 – 2026

- Delivered data and automation work for local clients alongside full-stack web projects in Python and Node.js.

Selected Projects

Demand Forecasting System *Python, PyCaret, scikit-learn, Docker, Streamlit*

- Sole author. Monthly forecasts across roughly 1,500 product series from a tiered zoo of about 150 labelled configurations, with one global model fitted across all series pooled rather than one per series, and a direct twelve-month horizon so error cannot compound.
- Scored out of sample on WAPE and MASE rather than MAPE, which explodes on zero months, with seasonal-naive as a mandatory baseline that ships if nothing beats it. Intervals are conformal, computed from model residuals rather than the variance of the data.
- A GREEN / AMBER / RED certification reports that median product-level WAPE is 16.6% and that 18.7% of series land inside the client's requested 10% band, rather than narrowing the bands until the requirement appears to be met.

Inventory & Sales Prediction *Python, LightGBM, FastAPI, PostgreSQL, Docker*

- Sole author. Three quantile LightGBM models give a lower, median and upper bound per product-store-day, over lag features at 1, 2, 3, 7, 14 and 28 days plus rolling mean, standard deviation, minimum and maximum windows and calendar features.
- Concept to production in one pipeline: train / validation / test split, WMAPE, MAE, RMSE and R^2 scoring, joblib persistence, a model loader that hot-reloads without a restart, a retrain entry point, and FastAPI serving in Docker.

StockSense AI *Python, FastAPI, LightGBM, Prophet, PostgreSQL, React, Streamlit*

- Item-level demand forecasting with LightGBM and Prophet, turned into inventory policy: safety stock, reorder points, EOQ and ABC classification, explored through interactive dashboards.

FinGraph AI *Node.js, Neo4j, Supabase pgvector, Hugging Face, Groq*

- Hybrid retrieval: MiniLM embeddings matched in pgvector above a similarity threshold, then every hit enriched with its Neo4j relationships before the answer is generated, so results carry the links behind them rather than the closest matching text.

Education & Ways of Working

B.E., Information Science and Engineering, Vivekananda Institute of Technology, Bangalore

Feature branches and conventional commits; GitLab CI with 132 pytest files as the merge gate on the voice-agent platform.